

VESSEL TYPES

BULK CARRIERS

Bulk carriers, or just bulkers, are ships especially designed to carry loose goods in bulk. The cargo transported in bulk commonly includes wood, coal, ore, grain, coke, fertilisers, cement, light minerals, sugar and sand.

Bulkers usually have one deck, with the engine room in the stern and a deckhouse above it. Holds are constructed with longitudinal and cross walls, called bulkheads. Cargo in bulk is easily stowed in between them. Bulk cargo ships are not equipped with any handling gear, except for handy size ones, up to 30,000 tons of deadweight. All loading and unloading is done by means of shore devices like grabs or suction pipes. Some of them make use of flexible ductings and fans, which simply blow light cargo into holds. Port devices may include special conveyors that drop cargo inside. When one hold is full, loading is continued into the next one.

Bulk carriers have large upper and lower ballast tanks to provide enough draught. Some bulk carriers are designed to function also as tankers. Such vessels are called Ore Bulk Oil (OBO) carriers.

CABLE-LAYING SHIPS

Cable-laying vessels, also called cable layers, are specially designed for laying and repairing telegraph and telephone cables across vast water areas like channels, seas and oceans. Modern cable layers are as efficient in repair and maintenance operations as in long-haul cable laying.

The internet boom together with the extraordinary expansion of telecommunication has led to the growing demand for vessels specialising in laying sub-sea optical fibre networks.

“The Cable Innovator” seems to be the largest vessel operating in this market. The ship was built by Kvaerner Masa Yards in Finland. All cable-laying operations are carried over the stern, so the vessel can maintain a high speed and is not slowed down during cable work. Moreover, it can operate successfully in extreme weather

conditions. It has been designed to deploy a remotely operated vehicle (ROV). The vehicle is connected to the ship via communication tether.

The most important cable handling equipment aboard "The Cable Innovator" is the electrically operated cable laying drum with various tension devices. The drum diameter is 4m long and has fixed-angle fleeting rings and blades for controlling the cable work. It is equipped with a special A-frame for handling the plough used for cable burial in the seabed.

Furthermore, "The Cable Innovator" is equipped with an echo sounder and devices for measuring the length of the cable laid out. While laying cables, all main data are monitored, logged and printed out as a quality control. The control system can also display all data accounting for the tension of the plough tow cable when the vessel operates and can activate the alarm in case the cable tension gets too high. The vessel automatically reduces its speed. When buried safely beneath the sea bed, the fibre optic communication cables constitute a vital part of our global telecommunication network.

CAR CARRIERS

The "Elbe Highway" is the first of the series of four innovative PCTC (Pure Car and Truck Carrier) Ro-Ro car carriers built in Gdynia Shipyard S.A., Poland, for long-term charter for Kawasaki Europe. The shipyard design office team initiated and accomplished all the concept work on these new vessels. Along the way they were able to put into practice the yard's considerable experience in the field.

The delivery ceremony was held on 20th August 2005, exactly nine months after the commencement of steel cutting. The second ship, the "Thames Highway", exactly replicating the design of the prototype, was completed by the end of 2005.

The largest car carriers of today can handle over 6000 units. The "Elbe Highway", with her overall length (L_{OA}) of 143 m and breadth (B) of 25 m, can carry up to 2100 units, so the total car deck capacity is comparatively small. In fact, she can be classified as the only vessel ever designed and built expressly for carrying vehicles. The intention was to follow the innovative approach to world car transport. The crucial

idea is to lower the cost of the port stay by means of employing smaller cargo ships, operating as feeders.

The vessel is constructed with two hydraulically-operated external stern ramps. One is a straight stern ramp, another one is a quarter stern ramp. Each of them has a safe working load of 70 tons and a 6-metre long driveway. All vehicles ranging from passenger cars to heavy movable machinery can be loaded and discharged through these two ramps.

For PCTC carriers, a stern quarter ramp offers considerable advantages for cargo access and handling. Its main benefit is that it allows the vessel to berth in the normal manner, that is alongside the quay, without the need for special shore facilities. The motor vehicles carried by the “Elbe Highway”, and later on by her three sisters, can be handled at any port in the world, not only at Ro-Ro terminals with trailer quays.

The carrier is built with eight cargo decks in total. Two of them are liftable car decks. They enhance greatly the operational flexibility of the vessel. Unlike hoistable car decks, they have no integral lifting mechanism, but are deployed by a mobile deck lift. Thus, the maintenance requirements of each of the panels forming the movable decks are limited.

The arrangement of internal ramps and fixed car decks follows the main design concept to shorten the time in port. Namely, deck supporting pillars have been designed off the ship centre line, making the construction not symmetrical, but allowing for better cargo access.

The “Elbe Highway” has been classified by DNV as ICE-1A as the hull is strengthened in her underwater and bow sections. The navigating bridge is fully glass-shielded to improve the navigation visibility on icy waters.

The prototype carrier has already joined the growing fleet of high-quality car carriers of high manoeuvrability and efficiency that are operating on short routes in the Baltic and North Sea regions.

CONTAINERSHIPS

These ships carry cargo in containers. Goods are locked and sealed in huge boxes of standard size. Containerships carry containers both in holds and on the main deck. In the holds, there is a special cellular structure of guide rails where containers are stowed one on the top of another. That is why they can be also called cellular vessels. These ships usually have one deck, with the machinery spaces located towards the aft end. Additional containers are stowed on open deck and anchored in place by wire ropes.

Containerships have produced a revolution in water transport. The higher speed of around 26 knots is their main advantage over other cargo ships. In addition, the loading and unloading work with the use of shore based moving gantry cranes is extremely fast. The primary advantage of the use of containers is the possibility of transporting cargo directly from customer to customer, not only from port to port.

Container vessels have grown in capacity up to 8000 TEU. Large container vessels usually do not have their own loading gear. However, small, or medium-sized ones, called container feeders, are often equipped with cargo gear. Also, some multi-purpose ships can operate as container feeders.

THE “ESTONIA” FERRY

The “Estonia” ferry was delivered in 1980 to operate on routes in the heavy traffic between Finland and Sweden. The ship was, at the time of delivery, the second largest ferry working in the Baltic Sea region. It was built with one bow ramp on the car deck, enclosed by a hinged bow visor that opened upwards, and two stern ramps. Passenger entrance doors were arranged on decks 4 and 5 and the pilot and bunkering doors located on the car deck.

The design of both the visor and ramp were very popular at the time. The installation included a bow visor and a loading ramp. The ramp was hinged at car deck level and was closed when in a raised position. In a closed position, the upper end of the ramp was extended into a box-like housing on the forecastle deck. The only reason for such a construction was to make space for the ramp when the visor was in its closed position.

The ramp was placed behind the bow visor. Thus, as you can see in the drawing above, the ramp was longer than the available height of the deck. There was only one reason for such a long ramp. Namely, the bulbous bow had become so long that the ramp needed to be extended to reach the quay edge.

While operating on the Baltic Sea on the 28th of September 1994, at about 0115 hrs, the visor separated from the bow and tilted over the stern. The ramp was pulled fully open, allowing vast amounts of water to enter the car deck. Consequently, the ship listed heavily to the starboard side. Many of the passengers were trapped in their cabins, with no chance to get out in time. A few minutes later, all four main engines stopped and the list increased as the water started to enter the accommodation area. Flooding of the ferry continued so fast that the starboard side submerged ten minutes later. The ship was sinking rapidly and disappeared from the radar screen at about 01.50hrs.

The alleged cause of water coming into the car deck resulting in the Estonia catastrophe was the poor construction and bad maintenance of the visor and bow ramp, together with too high operating speed. However, there were some other theories presented. According to one of them, the problem was that the bow visor

was placed in such a position that it could not be seen from the bridge. The bridge crew would have probably reacted if they had been able to observe the visor.

Another theory is based on the fact that the Estonia had not met the requirement of having an extra collision bulkhead, which should have been placed at more than 5% of the ship's length from the forward perpendicular. This would have definitely increased her chances of surviving the loss of the visor. If the collision bulkhead had been there, it would have prevented water from entering the car deck.

So the attempts to build vessels strong enough to restrain the sea have once again been conquered by the forces of the nature.

Study the following pairs of sentences and note that the second sentence in each pair has been transformed from the first one.

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|--|--------------|
| 1. The Finnish Company named the ferry "Viking Sally" | active form |
| The ferry was named "Viking Sally". | passive form |
| 2. The shipyard built the Estonia in 1980. | active form |
| The Estonia was built in 1980. | passive form |
| 3. People presented other theories. | active form |
| Other theories were presented. | passive form |

Note how it has been done and how the meaning has been changed.

The three pairs of sentences illustrate the basic usage of the passive.

- 1. The focus is on the activity itself; the message is about the change of name, not about the company.**
- 2. It is commonly known who builds ships, so this information can be left out; the focus is on the ship and the time of its building.**
- 3. The word "people" does not refer to specific individuals. It is impersonal. Everybody knows that only people define theories and can present them.**

The passive voice is constructed in the following way:



Education and Culture

Leonardo da Vinci

“TO BE” in the proper tense or verb structure	PAST PARTICIPLE form of the Verb
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INDUSTRIAL SHIPS

Industrial ships are designed to carry out industrial processes at sea, like drawing out raw materials and food resources from waters. The activities that take place aboard these ships include extracting oil and mineral salts, or catching and processing fish and crustaceans like crabs, shrimps and lobsters. Thus, if we consider the function the ships perform, we can clearly distinguish between the extractive and processing ships. The first type includes trawlers and seiners

A trawler is the most popular vessel among fishing ships. Its name comes from the name of the activity, 'trawling', which means catching fish by dragging a fishing net along the sea bed. The trawl can be launched either over the ship side or over the stern.

Spain and Norway have been taking the lead with respect to the number, size and the variety of trawlers built.

Non-trawling vessels can range from simple crafts which deploy a net, to fishing vessels that first lay out nets, even for a distance of a few kilometres , and then wait for the shoal of fish to swim into it.

The typical representatives are seiners, tuna clippers and crab boats.

A seiner makes use of a special kind of net called a seine net in the following way: the net hangs vertically in water. Its top edge floats and its bottom is weighted and equipped with a rope. When a shoal of fish swims into the net, the rope tightens and closes around it.

Processing ships receive fish or other sea goods from extractive ships, process them into products, and bring them to ports. They must be fitted with special machines for processing, canning and storing.

Pay attention to the word forms with **-ing** ending. You can easily notice that all of them have been highlighted in the reading above.

All the underlined word forms can fall into the following grammar groups and can function:

A. as a part of any continuous tense form

Examples:

- He is fishing.
- They will be catching whales.
- The ship has been sailing for three months.

B. as an adjective, describing causes of people's feelings and features of things and objects.

Examples:

- Racing yachts can sail very fast.
- Titanic was an amazing ship.
- Fishing boats are very popular in Iceland.

C. as a gerund, which is a noun derived from a verb and describes an activity and always has the same function as a noun (although it looks like a verb), so it can also be used as a subject of a sentence.

Examples:

- Hunting whales is wrong.
- Catching sea mammals is prohibited in some areas.
- Commercial fishing must be legal.

TANKERS

Tankers are vessels designed for carrying any liquid cargo such as petroleum and products derived from it, liquefied gases, chemicals, wine and water. There are gas tankers designed for carrying liquefied gas, either LPG or LNG, both of which need to be kept at higher pressure and at low temperatures to maintain the cargo in a liquefied state, and there are crude oil tankers. The latter usually carry crude oil from a loading port near oil fields or from the end of a pipeline to a refinery.

Gas tankers are often steam turbine ships. The boil-off, which is the gas evaporated from the cargo in order to keep the temperature low, can be used as fuel for the boilers.

Tankers come in all sizes, ranging from bunkering tankers of 1000 DWT used for refueling larger vessels to the real giants:

- the VLCC - Very Large Crude Carrier , 200,000 - 300,000 DWT
- the ULCC - Ultra Large Crude Carrier , over 300,000 DWT

Crude oil tankers are the largest of all cargo ships. Their capacity has risen right up to 500,000 tons and, consequently, their large draught limits their sailing routes. There are only a few ports that supertankers can enter and thus they are mostly loaded and unloaded from off-shore pumping stations. The liquefied cargo is loaded by means of pipes from shore facilities and through flexible pipelines mounted on the jetty.

A further step in the development of the oil industry is the Floating, Production, Storage and Offloading vessel (FPSO), designed for off-shore purposes.

When a large vessel like a crude oil tanker is damaged by collision or grounding, vast amounts of oil may leak out straight into the sea. This explains the strict requirement for them to have a double hull.